

Highlights

When does a trend–event decomposition help electricity price forecasting? Dual-field models with market-operator forecasts and quarterly recalibration

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- A trend–event dual-field model is adapted from reconstruction to price forecasting.
- On a static split the decomposition loses to a linear model with the same inputs.
- Quarterly refits and operator forecasts make the dual fields improve accuracy.
- The model gives the best probabilistic forecasts on negative-price hours.
- Ablations, Diebold–Mariano tests and error diagnoses trace each gain to a cause.

When does a trend–event decomposition help electricity price forecasting? Dual-field models with market-operator forecasts and quarterly recalibration

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Abstract

Wholesale electricity prices in markets with high renewable shares combine a slowly moving level with sparse events: scarcity spikes and, increasingly, negative-price troughs. Decomposing a series into a continuous trend and discrete events, as reconstruction models such as DualTimesField do, is a natural prior. We ask whether it improves day-ahead forecasting, and when. We adapt the dual-field structure to forecasting the next 24 hourly prices in three regions of the Australian National Electricity Market (2023–2024 test period, 17,521 origins per region) by detecting price events explicitly, adding the fields as a residual to a linear base, and conditioning on point-in-time forecasts from the market operator. Fitted once on 2015–2021, the decomposition does not help: a linear model with the same inputs is more accurate, because the fields add no information about the future and overfit a price regime that shifted after 2021. Two changes reverse this. Refitting before each test quarter removes most of the regime shift, and the operator’s predispach forecasts add information that no other input carries. The dual fields then reduce both the mean absolute error (MAE) and the continuous ranked probability score of the same linear base, and the model outperforms PatchTST, DLinear and a linear model (Diebold–Mariano $p < 0.01$). A per-hour calibrator and a validation-chosen price floor, derived from an error diagnosis, add significant gains. XGBoost with the same inputs keeps the lowest MAE (35.5 against 36.6 AUD/MWh), but our model gives the best probabilistic forecasts on negative-price hours of all models compared

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($p < 0.001$).

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1. Introduction

Short-term electricity price forecasts drive bidding, storage dispatch, demand response and hedging decisions [1, 2]. In markets with a fast-growing share of wind and solar generation, these forecasts have become harder. Prices still follow a daily and seasonal level, but they are increasingly punctuated by events: scarcity spikes when supply margins tighten and negative-price troughs when renewable output exceeds demand. In the Australian National Electricity Market (NEM), the share of hours with negative prices in Queensland rose from 1.6% in 2015–2021 to 13.2% in 2023–2024 (Section 3). A forecaster that treats these events as noise around a level misses exactly the hours in which storage and flexible demand earn or lose most.

Separating a series into a continuous trend and discrete events is therefore an attractive structural prior. Classical decompositions and their neural successors [3, 4, 5] split trend from seasonality; DualTimesField [6] goes further and represents a series as a continuous trend field (CTF), an implicit neural representation in continuous time [7, 8], plus a discrete event field (DGF) of localized atoms. It was designed for reconstruction and interpolation. Whether such a representation helps forecasting is not obvious: describing the past more faithfully does not by itself add information about the future, and a richer description of a past regime can overfit it.

This paper studies that question for day-ahead price forecasting in three NEM regions, under a same-data protocol in which every compared model receives the same inputs, the same training periods and the same evaluation. Our findings are sequential, and the paper follows them:

1. *Transplanted, the dual field does not separate events.* The CTF absorbs spikes and the DGF collapses. Detecting events explicitly, as the largest departures from the window median, restores the separation (Appendix A).
2. *Separation alone does not improve forecasts.* On a single fit to 2015–2021, a linear model on the same inputs is more accurate than the dual-field model, and adding the fields to that linear model makes it

worse (Section 6.2). The fields lower the training loss but raise the error on the 2022 validation year: they fit a price regime that did not persist.

3. *Recalibration and market-operator information make the fields useful.* Refitting every model before each test quarter, as is standard practice in electricity price forecasting [9], removes most of the regime shift. AEMO’s predispatch, the operator’s own dispatch simulation with current offers, adds forward-looking information that no other input carries and lowers the error of every model. In this setting the dual fields improve both point and probabilistic accuracy over the same linear base.
4. *Diagnosis-driven corrections.* A per-region analysis of the errors that remain against the strongest competitor locates them in negative-price hours, where linear heads over-extrapolate in the transformed target space. A shared per-hour calibrator and a price floor chosen on validation data address this and are each significant.

The final model is more accurate than PatchTST [10], DLinear [5] and a linear model, and gives the best probabilistic forecasts on negative-price hours of all compared models. XGBoost [11] with the same inputs and refits remains more accurate on mean absolute error, which we report and analyse rather than tune away. Beyond the model, the study offers a protocol lesson: a structured forecaster should be evaluated against strong same-input baselines and under the recalibration regime it will be deployed in, because both change which model is best.

The rest of the paper is organized as follows. Section 2 reviews related work. Section 3 describes the data and the forecasting task, Section 4 the model, and Section 5 the experimental protocol. Section 6 reports the results, Section 7 discusses them, and Section 8 concludes.

2. Related work

Electricity price forecasting. Reviews by Weron [1] and Nowotarski and Weron [12] survey point and probabilistic electricity price forecasting. Lago et al. [9] established an open benchmark and best practices, among them daily recalibration on a rolling window, variance-stabilizing transformations [13], and comparison against well-specified linear (LEAR) and neural baselines with Diebold–Mariano tests [14]. High-dimensional linear models [15] and distributional neural networks [16] are strong references. We follow these

67 practices: an asinh transformation, rolling recalibration, same-input baselines,
68 and significance tests.

69 *Deep time-series forecasting and decomposition.* Transformer forecasters such
70 as Informer [17], Autoformer [3], PatchTST [10] and iTransformer [18], re-
71 current models [19, 20] and DeepAR [21] have been widely applied. Zeng
72 et al. [5] showed that a decomposed linear model (DLinear) is competitive
73 with many of them. N-BEATS [4] and Autoformer use trend and seasonal
74 decompositions inside the network; hybrid models combine a linear component
75 with a nonlinear residual [22]. DualTimesField [6] represents trend and events
76 as continuous-time fields for reconstruction. Our model adds dual fields as a
77 residual to a linear base, and asks when this structure pays off in forecasting.

78 *Exogenous information and strong tabular baselines.* Fundamental forecasts
79 (load, renewable output, capacity) are the most informative exogenous inputs
80 in electricity price forecasting [1, 9]. Recent work adds text: Chen et al. [23]
81 extract market events from news with a large language model. We instead
82 use the market operator’s own forward-looking forecasts, the primary source
83 on which much market commentary is based. On tabular inputs, gradient-
84 boosted trees remain hard to beat [24], which is why XGBoost is a central
85 baseline here.

86 **3. Data and forecasting task**

87 *Market and prices.* We use hourly regional reference prices (RRP) and op-
88 erational demand for New South Wales (NSW1), Queensland (QLD1) and
89 Tasmania (TAS1) from AEMO’s aggregated price and demand data [25], 1
90 January 2015 to 31 December 2024 in market time (AEST), with half-hourly
91 records averaged to hours. Table 1 summarizes the series. The test period,
92 2023–2024, differs markedly from 2015–2021: negative prices are four to eight
93 times more frequent, spikes above 300 AUD/MWh are more frequent in NSW1
94 and QLD1, and 2022, the energy-crisis year that separates them, has a much
95 higher level.

96 *Forecasting task.* At every hour t of the test period, the task is to forecast
97 the prices of the next 24 hours, $p_{t+1}, \dots, p_{t+24}$, from information published
98 at or before t . Origins are hourly, so each region has 17,521 test origins. All
99 models see a 72-hour history of price and demand.

Table 1: Hourly regional reference prices (AUD/MWh) by period. Neg.: share of hours below zero; > 300: share of hours above 300 AUD/MWh.

Region	Period	Mean	Median	Std. dev.	Neg.	> 300
NSW1	2015–2021	70.5	58.0	164.9	0.37%	0.25%
	2022	182.7	135.4	215.3	2.61%	14.79%
	2023–2024	113.5	82.7	408.8	6.20%	1.75%
QLD1	2015–2021	70.9	55.2	202.5	1.63%	0.68%
	2022	205.1	139.9	456.2	4.05%	15.73%
	2023–2024	101.1	81.3	289.2	13.21%	1.85%
TAS1	2015–2021	69.4	57.6	68.7	2.53%	0.43%
	2022	147.0	104.9	285.3	1.82%	8.57%
	2023–2024	75.8	60.3	168.2	7.36%	0.50%

100 *Known inputs.* Four groups of forward-looking information enter every model,
101 each as published at the origin:

- 102 • *Calendar:* hour of day, day of week and day of year (sine and cosine),
103 month, weekend flag and a standardized year.
- 104 • *PD PASA:* AEMO’s pre-dispatch projected assessment of system ade-
105 quacy [26], from which we take maximum spare capacity and net load
106 (50%-probability demand minus semi-scheduled wind and solar avail-
107 ability) for each forecast hour, from the latest run published by the
108 origin, with the short-term PASA run filling hours beyond its horizon.
- 109 • *Gas:* the logarithm of the seven-day trailing mean of the Victorian
110 declared wholesale gas market price [27], using only gas days ended
111 before the origin.
- 112 • *Predispatch:* AEMO’s predispatch re-solves the market dispatch every
113 30 minutes with the current offers, constraints and demand forecasts,
114 and publishes regional prices and quantities up to the end of the next
115 trading day [26]. For each origin we take the latest run whose rows
116 were all published by then and keep the forecast price, total demand,
117 available generation and net interchange for each hour. Runs published
118 between about 04:00 and 12:30 end before the 24th forecast hour; the
119 uncovered hours repeat the last covered hour and are flagged (93% of

hours are covered). The archive has no run history for October 2022, whose origins are marked uncovered and filled with training-period medians.

Used directly as a forecast, the predispatch price has a test MAE of 147.5, 153.3 and 69.4 AUD/MWh in NSW1, QLD1 and TAS1, inflated by spikes that do not materialize, but a median absolute error of only 17–20 AUD/MWh: it is informative but must be calibrated.

Price treatments. We evaluate on raw prices and, as a robustness check that limits the weight of a few extreme hours in absolute-error metrics, on prices capped at 650 AUD/MWh, with the cap applied to the price series used as target and as history, in training and in scoring.

4. Method

Fig. 1 gives an overview. The forecast is the sum of a linear base on all inputs, a dual-field residual built from a trend–event decomposition of the history, and a calibration term, followed by a price floor. Every component is refit before each test quarter.

4.1. Target transformation

Prices are mapped by the area hyperbolic sine with median and median-absolute-deviation scaling [13],

$$y = \operatorname{asinh}\left(\frac{p - \mu}{\sigma}\right), \quad \mu = \operatorname{med}(p^{\text{train}}), \quad \sigma = \frac{\operatorname{med}|p^{\text{train}} - \mu|}{0.6745}, \quad (1)$$

and standardized with training statistics. The point forecast is produced in this space and mapped back by the inverse transform. The quantile forecasts are produced directly in standardized price space, which keeps their spread in the units on which they are scored.

4.2. Dual-field decomposition of the history

Let $x \in \mathbb{R}^{72 \times 2}$ be the standardized history of transformed price and demand, indexed by $\tau = 1, \dots, 72$.

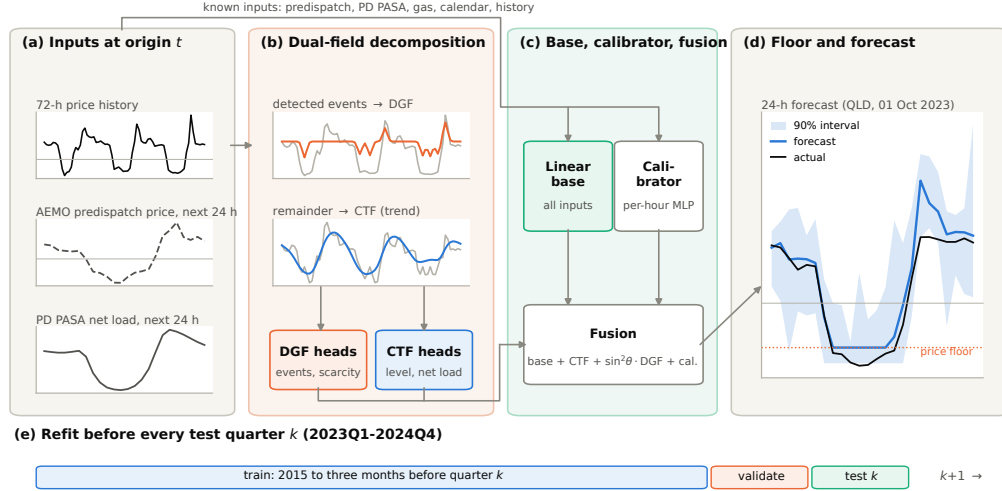


Figure 1: Overview. (a) At origin t the model reads the 72-hour history and the forecasts published by then (AEMO predispach, PD PASA, gas price, calendar). (b) Events detected in the history form the discrete event field (DGF); the continuous trend field (CTF) fits the remainder under a decomposition loss; field-specific linear heads read each field with matching inputs. (c) A linear base on all inputs and a per-hour calibrator are fused with the field heads. (d) The forecast is clipped at a price floor chosen on validation data. (e) Every model is refit before each test quarter. Thumbnails show one QLD1 test day (1 October 2023).

146 *Event field (DGF).* Events are detected in the price channel $x^{(1)}$ relative
 147 to a robust level. With departures $d_\tau = x_\tau^{(1)} - \text{med}_\tau x_\tau^{(1)}$, we select $K = 8$
 148 positions $c_1, \dots, c_K$ greedily by largest $|d_\tau|$, suppressing positions closer than
 149 three hours to an already selected one. Each event keeps its departure,
 150 soft-thresholded by a learned level $\kappa > 0$, and becomes a Gaussian bump of
 151 learned width w :

$$e(\tau) = \sum_{k=1}^K a_k \exp\left(-\frac{(\tau - c_k)^2}{2w^2}\right), \quad a_k = \text{sign}(d_{c_k}) \max(|d_{c_k}| - \kappa, 0). \quad (2)$$

152 Positions are data, not parameters, so the event field is fixed by the history;
 153 this is what makes the two fields separate (Appendix A).

154 *Trend field (CTF).* The CTF is a Fourier-feature implicit neural representa-
 155 tion [8, 6]. With normalized time $\tau' \in [0, 1]$, learned band-limited frequencies

156 $b_j = \omega_{\max} \text{sigmoid}(\beta_j)$, $j = 1, \dots, 16$, and a low-pass filtered copy $\tilde{x}_\tau$ of the
 157 history,

$$\gamma(\tau') = \left[\cos 2\pi b_j \tau', \sin 2\pi b_j \tau' \right]_{j=1}^{16}, \quad c(\tau) = f_\theta(\gamma(\tau'), \tilde{x}_\tau), \quad (3)$$

158 where f_θ is a small multilayer perceptron. The decomposition loss $\mathcal{L}_{\text{dec}} =$
 159 $\|c + e - x\|_2^2 / |x|$ makes the CTF fit what the events leave, and $r = x - c - e$
 160 is the remainder.

161 4.3. Forecast heads, linear base and fusion

162 For each of the 24 forecast hours h , linear heads map field features to a
 163 point forecast in transformed space and to quantiles at the levels 0.05, 0.10,
 164 0.50, 0.90 and 0.95 in price space:

- 165 • *CTF heads* read the trend c , the remainder r , the calendar, the gas
 166 price and the PD PASA net load: the level and its fundamental drivers.
- 167 • *DGF heads* read the event field e , the calendar, the PD PASA spare
 168 capacity and the predispach inputs; the DGF quantile head also reads
 169 the shortfalls of spare capacity below its training 1%, 5% and 10%
 170 quantiles, a piecewise-linear scarcity response.

171 The two are fused by an additive trigonometric gate, $\hat{y}_h^F = \hat{y}_h^C + \sin^2(\theta_h) \hat{y}_h^D$,
 172 with $\theta_h = \frac{\pi}{2} \text{sigmoid}(g_h)$ and g_h linear in the fields, the calendar and the DGF
 173 inputs; each quantile level has its own affine rescaling of g_h , so tail quantiles
 174 can draw on the event field more than the median.

175 The *linear base* $\hat{y}^B$ is one linear map from the raw history and all known
 176 inputs to the 24 point and quantile outputs. The fused field forecast is added
 177 to it as a residual, so the fields only need to model what a linear function of
 178 the inputs cannot.

179 *Calibrator*. A single two-layer perceptron (32 hidden units), shared by all
 180 forecast hours, reads hour h 's known inputs (predispach, PD PASA, calendar),
 181 a learned four-dimensional horizon embedding and the current price level
 182 (last observed value and 24-hour mean), and outputs additive corrections δ_h
 183 to the point and quantile forecasts. Its output layer is initialized at zero. It
 184 lets the model learn a nonlinear response to predispach, for example that
 185 high predispach prices are often not realized. The forecast before the floor is

$$\hat{y}_h = \hat{y}_h^B + \hat{y}_h^F + \delta_h. \quad (4)$$

186 *Training.* The parameters are trained jointly by minimizing

$$\mathcal{L} = \underbrace{\frac{1}{24} \sum_h (\hat{y}_h - y_h)^2}_{\text{point}} + \underbrace{\frac{1}{24.5} \sum_{h,q} \rho_q(\tilde{p}_h - \hat{\tilde{p}}_{h,q})}_{\text{quantiles}} + \mathcal{L}_{\text{dec}} + \lambda_s \mathcal{L}_{\text{smooth}} + \lambda_g \mathcal{L}_{\text{sparse}}, \quad (5)$$

187 where ρ_q is the pinball loss [28], $\tilde{p}$ is the standardized price, $\mathcal{L}_{\text{smooth}}$ penalizes
 188 first differences of the CTF, and $\mathcal{L}_{\text{sparse}}$ penalizes the share of active events,
 189 ten times more steeply above 30% ($\lambda_s = \lambda_g = 10^{-3}$). We use AdamW with
 190 learning rate 3×10^{-4} , weight decay 10^{-4} , batch size 128 and 30 epochs,
 191 and keep the epoch with the lowest validation loss. The model has 277,747
 192 parameters.

193 4.4. Price floor

194 Negative prices in the NEM are bounded in practice by renewable offers
 195 priced near the negative value of the associated certificates, but a linear head
 196 extrapolating in asinh space can, after the inverse transform, forecast prices far
 197 below any observed level (Section 6.3). We therefore clip forecasts from below,
 198 $\hat{p}_h \leftarrow \max(\hat{p}_h, \phi)$. For each refit and region, ϕ is chosen on that refit’s valida-
 199 tion quarter from $\{\text{none}\} \cup \{\min(Q_\alpha(p^{\text{train}}), 0) : \alpha \in \{0.05, 0.1, 0.2, 0.5, 1\}\%\}$,
 200 separately for the point forecast (by validation MAE) and the quantiles (by
 201 validation CRPS). The test quarter is never used.

202 4.5. Quarterly recalibration

203 Before each of the eight test quarters k (2023Q1 to 2024Q4), every model
 204 is refit on data from January 2015 to three months before the quarter, selects
 205 its checkpoint on those three months, and forecasts the origins of the quarter.
 206 Forecasts of the eight quarters together cover exactly the 17,521 test origins
 207 of each region. Transformations, standardizations and quantile knots are
 208 re-estimated on each refit’s training data.

209 5. Experimental setup

210 *Baselines.* All baselines receive the same history and known inputs, including
 211 predispatch, and are refit on the same quarterly schedule.

- 212 • *XGBoost* [11]: one multi-quantile model per forecast hour on the flat-
 213 tened history, gas price and that hour’s calendar and known inputs
 214 (depth 6, learning rate 0.05, subsampling 0.8, up to 2,000 rounds with
 215 early stopping on validation); the median is the point forecast.

- 216 • *PatchTST* [10] (patch length 16, stride 8, instance normalization, three
217 layers, $d_{\text{model}} = 128$) and *DLinear* [5] (moving-average decomposition).
218 Both are channel-independent and would otherwise ignore the known
219 inputs, so each adds a linear map from the known future inputs and
220 the demand history to its outputs.
- 221 • *Linear*: one linear map of the price history plus the same linear map of
222 known inputs, with a mean-squared-error point output and pinball-loss
223 quantiles; in our experiments it is the strongest linear specification and
224 dominates DLinear.
- 225 • *Linear base only*: our model without the dual fields, the calibrator and
226 the floor, trained by the same pipeline; it isolates the contribution of
227 the structure.

228 Neural baselines output the five quantiles of the transformed price and are
229 trained with the pinball loss (Adam, batch size 256, 30 epochs, weight decay
230 10^{-5} , best validation epoch); their learning rates were chosen from $\{10^{-4}, 5 \cdot$
231 $10^{-4}, 10^{-3}\}$ on validation data. Two further neural baselines, iTransformer
232 [18] and Informer [17], were less accurate than PatchTST and DLinear in a
233 static-split screening (Appendix B) and are not refit.

234 *Metrics.* Point forecasts are scored by MAE and by window RMSE, the
235 root mean squared error of each 24-hour window averaged over windows.
236 Probabilistic forecasts are scored by the coverage and average interval score
237 (AIS) of the 5–95% interval and by an approximation of the continuous ranked
238 probability score [29], twice the mean pinball loss over the five quantiles
239 (CRPS). MAE is minimized by the median [30]. We also score the tails:
240 CRPS and the 0.05-quantile pinball loss on hours with negative prices, and
241 CRPS and the 0.95-quantile pinball loss on hours above 300 AUD/MWh.

242 *Statistical tests.* Each model is trained with seeds 2026, 2027 and 2028; tables
243 report seed means. Differences are tested with the Diebold–Mariano test
244 [14] on per-origin losses (MAE averaged over the 24 hours and the seeds;
245 CRPS of seed-averaged quantiles), with a Newey–West long-run variance [31]
246 with 48 lags because hourly origins overlap. Pooled tests use the three-region
247 mean loss at each origin. For tail losses, each origin’s loss is its mean over
248 the event hours in its horizon, and origins without such hours are dropped.
249 Out-of-sample evaluation follows Tashman [32].

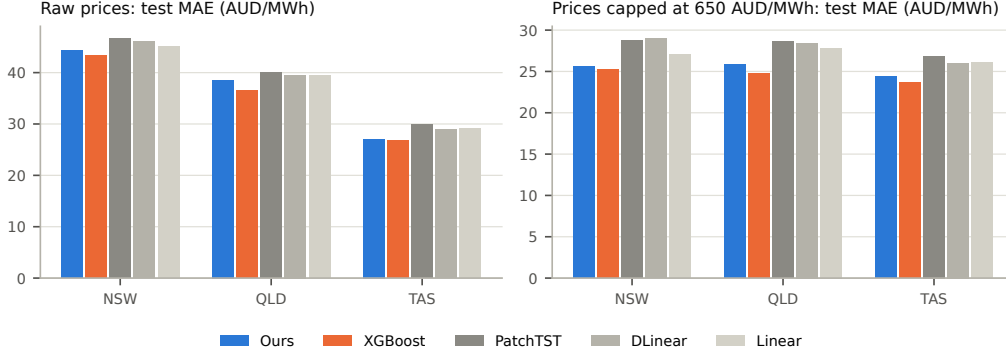


Figure 2: Test MAE per region in the main setting, raw prices (left) and prices capped at 650 AUD/MWh (right); means over three seeds.

250 *Implementation.* The models are implemented in PyTorch 2.5 and XGBoost
 251 3.2 and trained on NVIDIA L40 GPUs. Code and configurations reproduce
 252 every table and figure.

253 6. Results

254 6.1. Main comparison

255 Table 2 and Fig. 2 compare the models in the main setting, with quarterly
 256 refits and predispatch inputs, and Table 3 tests the differences. The dual-
 257 field model reduces the mean MAE of the same linear base from 37.60 to
 258 36.64 AUD/MWh on raw prices and from 26.72 to 25.29 on capped prices,
 259 and its CRPS from 26.04 to 25.07 and from 14.96 to 13.80 (all $p < 0.01$).
 260 It is more accurate than PatchTST, DLinear and the linear model on MAE
 261 under both price treatments and in every region ($p < 0.05$), and on capped
 262 CRPS ($p < 0.01$); on raw CRPS the differences to these three models are not
 263 significant. XGBoost has the lowest MAE and CRPS: our model is behind
 264 by 1.11 (raw) and 0.75 (capped) on MAE, most in QLD1 (2.10 raw); in TAS1
 265 the raw MAE difference is not significant.

266 6.2. When does the decomposition help?

267 Fig. 3 and Table 4 trace every model across three settings, and the first
 268 block of Table 5 isolates the dual fields against the same linear base in each.

269 On the static split (a single fit to 2015–2021), the fields *raise* MAE by 0.92
 270 (raw) and 0.95 (capped) ($p < 0.01$), and the dual-field model (41.54) is behind

Table 2: Main comparison on the 2023–2024 test set (17,521 hourly origins per region, 24-hour horizon). Every model is refit before each test quarter and uses the same inputs, including AEMO predispatch. Cells are means over three seeds; Mean MAE also shows the seed standard deviation. Ours includes the calibrator and the price floor; Linear base only is ours without them and without the dual fields. Lower is better; best in bold.

Model	MAE			Mean MAE	Window RMSE	90% AIS	CRPS
	NSW	QLD	TAS				
<i>Raw prices</i>							
Ours	44.27	38.57	27.07	36.64 ± 0.14	73.69	367.8	25.07
Linear base only	44.86	39.01	28.91	37.60 ± 0.03	74.35	401.2	26.04
XGBoost	43.34	36.47	26.76	35.52 ± 0.02	73.14	347.6	23.11
PatchTST	46.68	40.03	29.84	38.85 ± 0.07	75.94	355.3	24.33
DLinear	46.09	39.54	28.98	38.20 ± 0.09	76.11	338.7	23.62
Linear	45.16	39.51	29.10	37.92 ± 0.02	74.78	339.7	23.66
<i>Capped at 650</i>							
Ours	25.64	25.89	24.35	25.29 ± 0.17	36.07	165.8	13.80
Linear base only	26.77	27.46	25.92	26.72 ± 0.00	38.08	183.8	14.96
XGBoost	25.20	24.78	23.65	24.54 ± 0.02	35.51	160.7	13.10
PatchTST	28.77	28.67	26.87	28.10 ± 0.07	40.00	188.6	15.12
DLinear	29.01	28.44	26.02	27.82 ± 0.05	41.50	185.7	14.96
Linear	27.09	27.84	26.11	27.01 ± 0.01	38.42	186.1	14.98

the linear model with the same inputs (40.52). The fields lower the training loss but raise the error on the 2022 validation year: at the selected epoch in NSW1, validation MAE is 60.6 with the fields against 48.6 without. They fit the 2015–2021 level, which shifted in 2022 and again in 2023–2024 (Table 1). Normalizing each history window to remove the level did not help, because volatility shifted as well: the median window standard deviation of QLD1 prices is 2.2 times larger in the test period than in training (Appendix B).

With quarterly refits, the fields no longer cost point accuracy (-0.10 raw, -0.06 capped, not significant) and lower CRPS by about 1.1–1.3 ($p < 0.01$). With predispatch inputs as well, they lower both MAE (-0.34 raw, $p < 0.05$; -0.70 capped, $p < 0.01$) and CRPS (-1.13 , -0.94 , $p < 0.01$). Predispatch lowers the MAE of every model, and of XGBoost most (39.53 to 35.52 raw): it carries information about current offers that no other input has.

Table 3: Diebold–Mariano tests, ours minus comparator, in the main setting (negative favours ours). Δ columns pool the three regions; NSW, QLD and TAS give the raw-price MAE difference per region. Newey–West variance with 48 lags. $*p < 0.05$, $**p < 0.01$.

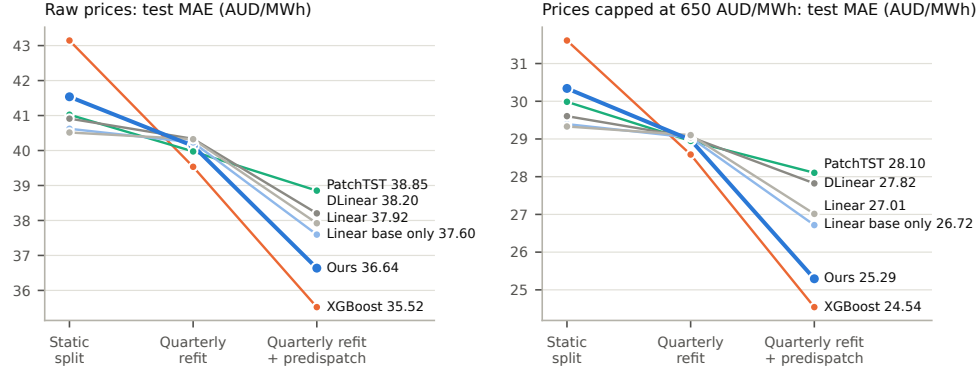
Comparator	Raw prices					Capped at 650	
	Δ MAE	NSW	QLD	TAS	Δ CRPS	Δ MAE	Δ CRPS
Linear base only	-0.96**	-0.59	-0.44	-1.84**	-1.42**	-1.42**	-1.18**
XGBoost	+1.11**	+0.93**	+2.10**	+0.31	+1.11**	+0.75**	+0.31**
PatchTST	-2.22**	-2.41**	-1.46**	-2.77**	-0.20	-2.81**	-1.69**
DLinear	-1.57**	-1.82*	-0.97*	-1.91**	+0.58	-2.53**	-1.46**
Linear	-1.28**	-0.89*	-0.93**	-2.03**	+0.53	-1.72**	-1.49**

Table 4: Three-region mean test MAE across settings: static split, quarterly refit, quarterly refit with predispatch (PD). Means over three seeds; best per column in bold.

Model	Raw prices			Capped at 650		
	Static	Refit	Refit + PD	Static	Refit	Refit + PD
Ours	41.54	40.13	36.64	30.34	28.98	25.29
Linear base only	40.62	40.23	37.60	29.39	29.04	26.72
XGBoost	43.15	39.53	35.52	31.61	28.59	24.54
PatchTST	41.02	39.97	38.85	29.99	28.95	28.10
DLinear	40.91	40.33	38.20	29.61	29.05	27.82
Linear	40.52	40.32	37.92	29.33	29.10	27.01

284 6.3. Diagnosis and corrections

285 To locate the remaining gap to XGBoost, we split the absolute-error
286 difference by condition. In QLD1, where the gap was largest (2.9 AUD/MWh
287 before the corrections), hours with a non-positive actual price (13% of hours)
288 account for 47% of it, and hours in the lowest net-load quintile, largely the
289 same midday solar hours, for 48%. The dual-field model forecast prices below
290 -100 AUD/MWh in 0.23% of hours, against 0.01% of actual prices and none
291 for XGBoost: the 3.2% of hours in which it forecast below -50 carry 31% of
292 the gap. Linear heads extrapolate in asinh space, and the inverse transform
293 turns small extrapolations into deep negative prices, whereas trees stay inside
294 the range seen in training.



Three-region means over seeds 2026-2028. Ours in the last setting includes the calibrator and the validation-chosen price floor. Axes do not start at zero.

Figure 3: Three-region mean test MAE of every model in three settings: a single fit on 2015–2021, quarterly refits, and quarterly refits with predispatch. Ours in the last setting includes the calibrator and the price floor. Means over three seeds; axes do not start at zero.

Two corrections follow (second block of Table 5). Letting the model respond nonlinearly to predispatch, by hinge features of the predispatch price or by the calibrator, lowers MAE by 0.25–0.44 ($p < 0.01$ except the raw calibrator, $p < 0.05$); the calibrator is retained because it also gives the best tail scores. The validation-chosen price floor lowers MAE by a further 0.31 (raw) and 0.29 (capped), significantly in every region and most in QLD1 (−0.82 raw). In QLD1 the chosen point floors are −37 to −48 AUD/MWh in seven of the eight refits. Adding predispatch interconnector flows and limits does not help (+0.04 raw, +0.19 capped): the predispatch price already reflects the network constraints.

6.4. Price tails

Table 6 scores the tails. On negative-price hours, our model has the lowest CRPS and 0.05-quantile pinball loss of all models under both price treatments, significantly lower than every comparator including XGBoost (raw CRPS 14.52 against 15.96; $p < 0.001$). On capped prices it also has the lowest CRPS on spike hours (121.1 against 138.3 for XGBoost; $p < 0.01$). On raw spike hours it ties XGBoost and is worse than PatchTST, DLinear and the linear model ($p < 0.05$), whose upper quantiles react more to extreme predispatch prices.

Table 5: Ablations. Pooled Diebold–Mariano differences (negative is better). The first block compares the dual fields with the same linear base in each setting; each row of the second block is compared with the model it modifies. * $p < 0.05$, ** $p < 0.01$.

Change	Raw prices		Capped at 650	
	Δ MAE	Δ CRPS	Δ MAE	Δ CRPS
<i>Fields minus linear base</i>				
Static split	+0.92**	-1.90**	+0.95**	-1.88**
Quarterly refit	-0.10	-1.28**	-0.06	-1.06**
Quarterly refit + predispach	-0.34*	-1.13**	-0.70**	-0.94**
<i>Added to fields + predispach</i>				
Hinge features on predispach price	-0.25**	-0.95*	-0.39**	-0.26**
Calibrator	-0.31*	-0.26*	-0.44**	-0.22**
Price floor (after calibrator)	-0.31**	-0.03**	-0.29**	-0.01
Interconnector inputs (after floor)	+0.04	+0.04	+0.19**	-0.04

314 6.5. Case studies

315 Fig. 4 shows two test days chosen by fixed rules. On the QLD1 day
316 with the lowest mean price (1 October 2023), the linear base and the dual
317 fields keep the midday forecast positive; the calibrator brings it into negative
318 prices but overshoots to about -85 AUD/MWh, and the floor (-43) clips the
319 overshoot. The day’s MAE is 12.2 for our model, 14.1 for XGBoost and 17.3
320 for predispach. On the NSW1 day on which predispach most overstated
321 the peak (24 October 2024, predispach 17,500 AUD/MWh against an actual
322 price below 300), every model is pulled up at 17:00; ours reaches about 470
323 and XGBoost about 280 (day MAE 57.1 and 52.8). Such phantom spikes
324 remain a failure mode.

325 6.6. Computational cost

326 Table 7 reports the size of each model and the wall-clock time of one
327 quarterly refit (NSW1, 2023Q3 test quarter, seed 2026) on one GPU. Every
328 model refits in four to six minutes; the dual-field model costs about 20% more
329 than its linear base and less than PatchTST. A full evaluation (eight refits,
330 three regions, three seeds) takes a few hours per model on one GPU.

Table 6: Probabilistic scores in the price tails, main setting, mean of the three regions, seed-averaged quantiles. Lower is better; best in bold. [†]/[‡]: ours is significantly better/worse than that model (pooled Diebold–Mariano test on per-origin event-hour losses, $p < 0.05$).

Model	Negative-price hours		Spike hours (>300)		
	CRPS	$q_{0.05}$ pinball	CRPS	$q_{0.95}$ pinball	90% coverage
<i>Raw prices</i>					
Ours	14.52	4.74	800.3	675.3	91.5%
Linear base only	16.21 [†]	5.82 [†]	882.3 [†]	777.2 [†]	93.1%
XGBoost	15.96 [†]	6.13 [†]	790.5	640.3	87.7%
PatchTST	17.99 [†]	7.51 [†]	745.2 [‡]	600.3 [‡]	90.0%
DLinear	17.25 [†]	7.33 [†]	740.3[‡]	591.7[‡]	89.1%
Linear	17.25 [†]	7.38 [†]	741.8 [‡]	593.8 [‡]	89.0%
<i>Capped at 650</i>					
Ours	14.28	4.83	121.1	77.2	90.5%
Linear base only	16.23 [†]	6.07 [†]	146.7 [†]	87.5 [†]	92.4%
XGBoost	15.47 [†]	6.13 [†]	138.3 [†]	91.0 [†]	86.2%
PatchTST	17.90 [†]	7.42 [†]	127.8	77.0	90.0%
DLinear	17.19 [†]	7.32 [†]	141.7 [†]	85.5	89.0%
Linear	17.31 [†]	7.44 [†]	141.9 [†]	85.5	88.7%

7. Discussion

Why decomposition alone does not help. Decomposing the history is a change of representation: $c + e + r = x$, so the heads receive no information that a linear map of x lacks. Its possible benefit is an inductive bias, and on a static split that bias works against the model, because the fields describe the price regime of the training period and it shifted. The same pattern appears in the case of Informer, which lacks instance normalization and overfit the 2015–2021 level in our screening (Appendix B).

Why recalibration and predispatch make it help. Refitting brings the training distribution close to the test quarter, so a richer description of recent history transfers. Predispatch adds information about the future that the history does not contain; the fields then have something to combine with, and the gated event heads respond to scarcity and to predispatch where a linear base cannot. The improvement is largest where events are frequent and

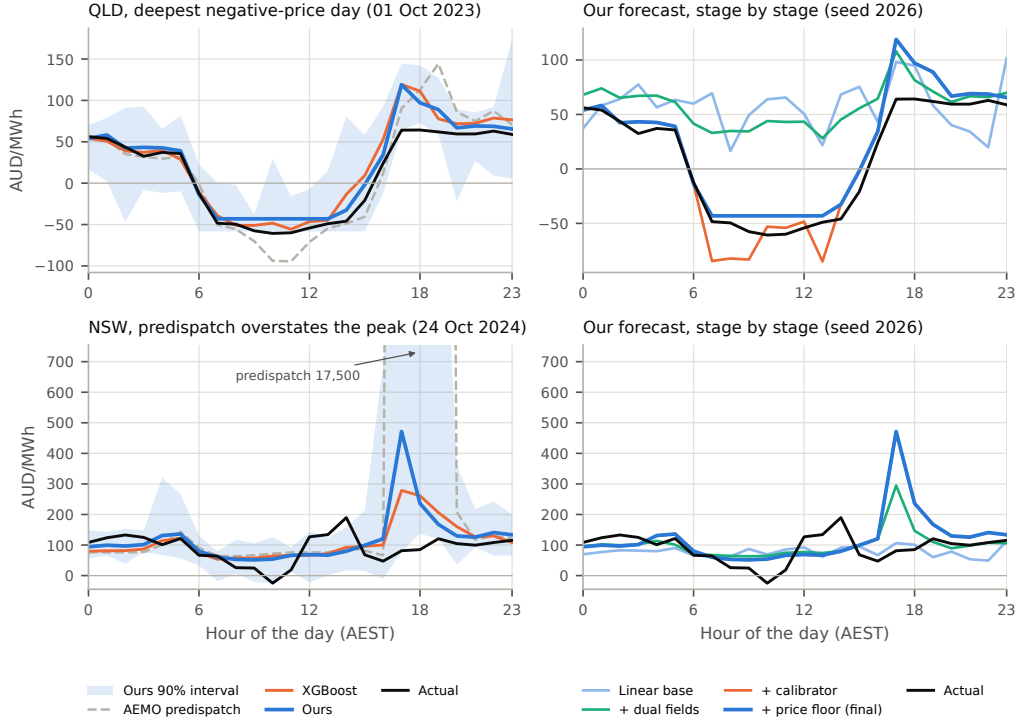


Figure 4: Case studies, chosen by fixed rules on the test split. Top: QLD1 day with the lowest mean price. Bottom: NSW1 day on which predispach most overstated the peak (forecast above 1,000, actual below 300 AUD/MWh). Left: our forecast with its 90% interval, XGBoost, predispach and the actual price. Right: our forecast stage by stage for the seed-2026 refit; on the NSW1 day the floor is not binding, so the calibrated and final forecasts coincide.

regular: negative-price hours, driven by midday solar output and visible in the predispach and net-load forecasts.

The gap to XGBoost. Trees use predispach better on average: they learn threshold responses, such as discounting high predispach prices, and cannot extrapolate beyond the training range. The calibrator and the floor recover part of this, but XGBoost remains ahead in the 100–300 AUD/MWh band and in QLD1. Practitioners who care about mean absolute error alone should prefer gradient-boosted trees on these inputs; the dual-field model is preferable when negative-price hours and calibrated tails matter, for example for storage and flexible demand.

Table 7: Model size and wall-clock time of one quarterly refit (NSW1, 2023Q3, seed 2026), each run alone on an NVIDIA L40 shared with other users; times are indicative. XGBoost size is the total number of boosting rounds over the 24 per-hour models; Linear base only keeps the field modules, which are still fitted by the decomposition loss.

Model	Parameters	Seconds per refit
Ours	277,747	312
Linear base only	275,629	258
XGBoost	2,871 rounds	269
PatchTST	599,730	364
DLinear	78,120	252
Linear	83,232	237

355 *Limitations.* The study covers three regions of one market and a two-year
356 test period. The price floor is a post-hoc clip chosen on validation data rather
357 than part of the trained model. On raw spike hours the model is behind the
358 linear models. Predispatch run history is missing for October 2022, and the
359 predispatch horizon does not always cover the 24th hour. Hyperparameters of
360 the dual-field model were developed on the static split; they were not re-tuned
361 for the recalibrated setting.

362 8. Conclusion

363 A trend–event decomposition helps electricity price forecasting only under
364 the right conditions. Adapted from reconstruction, the dual-field structure
365 separates price events once they are detected explicitly, but on a static
366 split it is less accurate than a linear model with the same inputs. With
367 quarterly recalibration and the market operator’s predispatch forecasts, the
368 same structure significantly improves point and probabilistic accuracy over its
369 linear base, is more accurate than PatchTST, DLinear and a linear model, and
370 gives the best probabilistic forecasts on negative-price hours of all compared
371 models; a calibrator and a price floor, motivated by an error diagnosis,
372 add further gains. XGBoost on the same inputs remains more accurate on
373 mean absolute error. Future work should move the price floor into training,
374 model the predispatch error explicitly to handle phantom spikes, and test the
375 approach in other markets with frequent negative prices.

376 **CRedit authorship contribution statement**

377 **zl:** Conceptualization, Methodology, Software, Validation, Formal analysis,
378 Investigation, Data curation, Writing – original draft, Writing – review &
379 editing, Visualization.

380 **Declaration of competing interest**

381 The authors declare that they have no known competing financial interests
382 or personal relationships that could have appeared to influence the work
383 reported in this paper.

384 **Declaration of generative AI and AI-assisted technologies in the**
385 **manuscript preparation process**

386 During the preparation of this work the author(s) used [tool name] in
387 order to [purpose]. After using this tool, the author(s) reviewed and edited
388 the content as needed and take(s) full responsibility for the content of the
389 published article.

390 **Data availability**

391 All data are public and were obtained from the Australian Energy Market
392 Operator [25, 26, 27]. Code and configurations that reproduce the results will
393 be made available.

394 **Appendix A. Field separation and static-split development**

395 Transplanted from reconstruction, the original DGF of DualTimesField,
396 with atoms at globally learned positions, collapses in forecasting: the CTF
397 absorbs the spikes, and the event field carries -0.04 to 0.00 of past spike-hour
398 prices on the test split. With detected events it carries 0.28 to 0.33 (Fig. A.5,
399 Table A.8). Fig. A.6 shows the development on the static split, where most
400 of the gain came from the target transformation and the fundamental inputs,
401 and Table A.9 the structural and detector ablations of the detected-event
402 DGF: suppression of nearby events and the median level matter, while the
403 number of events and the threshold matter little.

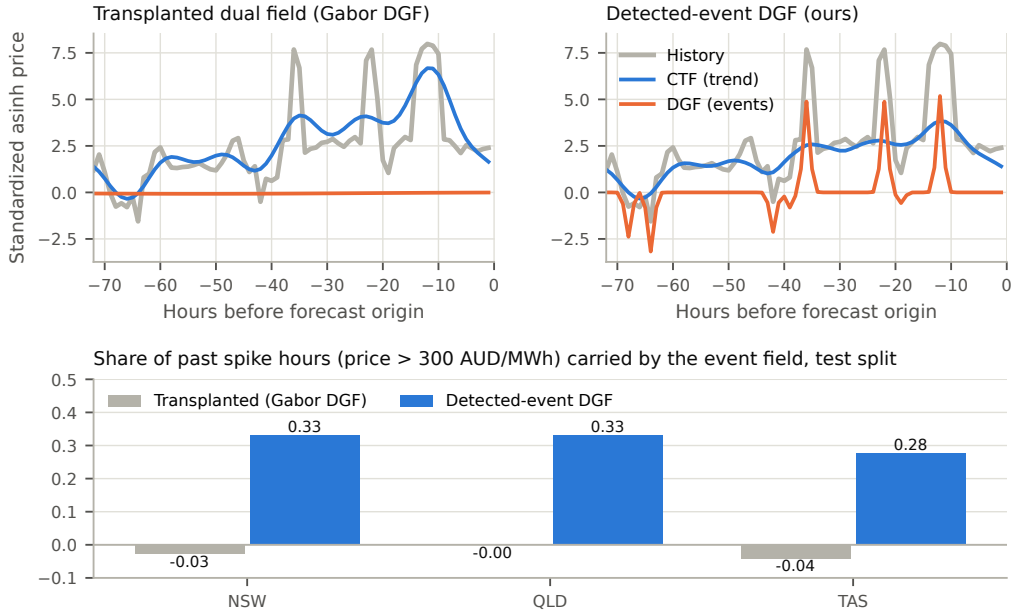


Figure A.5: Field separation. Top: one NSW1 test window in standardized asinh price; transplanted, the Gabor DGF stays flat and the CTF absorbs the spikes (left); with detected events the spikes and troughs move into the DGF (right). Bottom: share of past spike-hour prices (> 300 AUD/MWh) carried by the event field on the test split.

404 Appendix B. Designs that did not help

405 The following were tested with three seeds and are reported for complete-
 406 ness (full records are in the code repository).

- 407 • *Window-normalized fields* on the linear base, with the residual rescaled
 408 by the window's price scale: raw MAE 44.29 against 41.54 unnormalized
 409 (static split), because test-period volatility is about twice the training
 410 volatility.
- 411 • *Fields on the quantiles only*, with asinh-space quantiles and the point
 412 forecast from the linear base: the fields widened the 90% interval from
 413 171 to 418 AUD/MWh and raised CRPS from 28.94 to 31.25 (static
 414 split).
- 415 • *A future event field*, a three-component mixture for spike and trough

Table A.8: Share of past spike-hour prices carried by the event field on the test split.

Event field	NSW	QLD	TAS
Transplanted (Gabor)	-0.03	-0.00	-0.04
Detected events (ours)	0.33	0.33	0.28

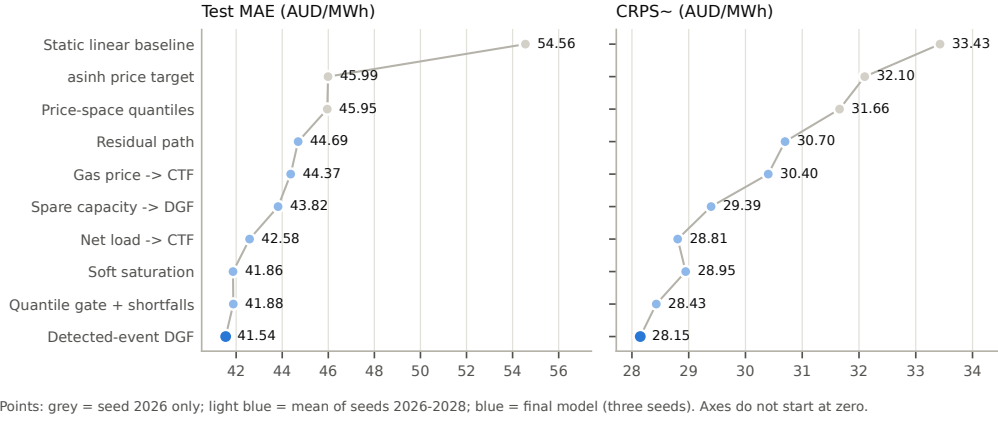


Figure A.6: Development on the static split: three-region mean test MAE and CRPS on raw prices; each step changes one component. Grey: seed 2026 only; blue: means over seeds 2026–2028. Axes do not start at zero.

- 416 probabilities, did not beat a logistic regression on the same inputs in
417 precision–recall.
- 418 • *iTransformer* and *Informer* (static split, raw MAE 43.20 and 46.65) were
419 less accurate than PatchTST (41.02) and DLinear (40.91); Informer’s
420 validation loss was lowest after one to five epochs.
 - 421 • *Predispatch interconnector inputs*: no gain (Section 6.3).

Table A.9: Structural and detector ablations of the detected-event DGF on the static split, raw prices, three-region means over three seeds (MAE $\pm$ seed standard deviation). The last column gives the paired MAE difference to the detected-event DGF per seed; positive means the variant is worse.

Variant	MAE	Window RMSE	CRPS	Δ MAE vs detected (seeds)
<i>Structure</i>				
Transplanted dual field (Gabor DGF)	41.88 $\pm$ 0.22	79.21	28.43	+0.38 / +0.03 / +0.58
No decomposition (heads read raw history)	41.90 $\pm$ 0.11	79.31	28.40	+0.30 / +0.25 / +0.51
Adaptive attention atoms + echo	41.84 $\pm$ 0.05	79.01	28.09	+0.30 / +0.28 / +0.30
Detected events + daily echo	42.21 $\pm$ 0.11	79.93	28.22	+0.79 / +0.47 / +0.73
Detected-event DGF	41.54 $\pm$ 0.06	78.80	28.15	–
<i>Detector settings (8 events, 3-hour separation, learned threshold, median)</i>				
4 events	41.65 $\pm$ 0.03	78.85	28.25	+0.13 / +0.01 / +0.18
16 events	41.50 $\pm$ 0.08	78.76	28.15	-0.08 / -0.03 / -0.03
No suppression (1-hour separation)	41.85 $\pm$ 0.03	79.00	28.05	+0.36 / +0.21 / +0.35
6-hour separation	41.51 $\pm$ 0.05	78.77	28.28	-0.05 / -0.11 / +0.05
No threshold	41.63 $\pm$ 0.08	78.98	27.97	+0.21 / -0.03 / +0.07
Mean baseline	41.64 $\pm$ 0.03	78.91	28.02	+0.09 / +0.05 / +0.15

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